Nhan Nguyen

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Technical Skills

Languages: Java, Javascript, Typescript, Python, C++, HTML/CSS, GDScript
Frameworks/Libraries: Flask, Next.js, React, Tailwind CSS, Express.js, Pandas, Scikit-learn, PyTorch
Technologies: Docker, DataGrip, MySQL, PostgreSQL, MongoDB, Figma, Microsoft Office, Azure, Supabase

Experience

Machine Learning Engineer **June 2026 – Present**
American Society of Mechanical Engineers *Long Beach, CA*

- Contributed in the construction of an autonomous glider for iAM3D through a Raspberry Pi 4-based computer vision pipeline using **OpenCV**-based visualization, distance measurement, gyroscope and GPS telemetry.
- Ensured accurate real-time object detection by training a **YOLOv8n** object detection model on a **15,000+** image custom **Pascal** and **COCO** based dataset using **ClearML** across **10** epochs with a batch size of **8** images with **6** classes, achieving **0.77** and **0.87** F1 scores on person and building detection.
- Enhanced autonomous flight analysis by integrating a ground control user interface with Python sockets, displaying bounding boxes, roll, pitch, and GPS telemetry with an average latency of **0.3** seconds.

Web Developer **August 2025 – June 2026**
Union of Vietnamese Student Associations of Southern California *Garden Grove, CA*

- Designed and developed a large-scale community platform from scratch through the **Agile** development framework, executing weekly sprint cycles to deliver iterative design and features under 1-2 week constraints.
- Implemented responsive front-end architecture using **TypeScript**, **React**, and **Tailwind CSS**, increasing cross-device compatibility and reducing page load times by up to **200%**.
- Opened and received approval for **15** pull requests concerning major features/issues, such as site-wide mobile responsiveness, addition of whole pages, and resolving component implementation or styling issues.

Frontend Developer **August 2025 – March 2026**
Association of Computing Machinery *Long Beach, CA*

- Contributed to development of front-end event infrastructure and registration systems using **JavaScript**, **Tailwind CSS**, and **GSAP**, processing **300+** applications and supporting seamless user experiences for **137** attendees.
- Designed and implemented a dynamic website section with animated GSAP components showcasing **6** hackathon tracks and **12** sponsors, meeting development goals within a 1 week constraint.

Education

California State University, Long Beach **Long Beach, CA**
B.S. Computer Science, Minor in Statistics *Expected Graduation: May 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Programming, Software Engineering, Operating Systems, Databases, Machine Learning, Artificial Intelligence

Projects

Desktop Job Sim **https://github.com/Detims/desktop-job-sim**

- Built a cross-platform desktop pet productivity app using **Electron**, **React**, **TypeScript**, and **Vite**, featuring draggable pet interactions, animated mood states, hover-based stat overlays, and transparent window states.
- Integrated **Google API** for **Gmail**, **Calendar**, and task management data, allowing the system to read recent emails, track upcoming events, monitor task activity, and convert productivity signals into in-app rewards.

Fire Detection System

- Developed an **IoT**-based fire detection system integrating flame, CO2, and smoke sensors, streaming real-time data to **NeonDB** for continuous monitoring and analysis of environmental risk factors.
- Engineered a data pipeline with **Pandas** and **Scikit-learn** to preprocess and clean a high-frequency sensor dataset of **50,000+** records, utilizing moving averages with rolling windows to enable accurate feature engineering and training.
- Built and optimized a **Random Forest classifier** (max_depth=8, max_features=12), achieving **96%** accuracy, **90%** precision, and **0.67** F1 score, improving predictive analytics over **Logistic Regression** and **Support Vector Machines**.

InboxAssist **https://github.com/magichouse21/InboxAssist**

- Collaborated in the construction of a **Azure**-based Google extension for Outlook, utilizing **LLM API** calls to enable email summarization, smart search, Q&A, assisted drafting, and sending emails.
- Implemented Microsoft **OAuth** and local state management, allowing browser-based sign-in and persistent sessions.
- Improved email parsing techniques for unread email summarization using the **Microsoft Graph API** with body_preview attribute, resulting in a **87%** decrease in token utilization.
- Replaced legacy **FAISS**-based retrieval architecture with Javascript-based chunking and vector embeddings.

Discord Bot **https://github.com/Detims/discord-bot**

- Developed a **Python-based** application integrated with the Discord API, utilizing **PostgreSQL** through **Supabase** to store message information, user scores, and update server leaderboards. Deployed with a **Docker** image for scaling self-hosting.
- Involved **NLTK** for sentiment analysis and **Google Gemini** / **OpenAI GPT APIs** for intelligent responses.